Python and Ruby. This platform powers over 75,000 mobile applications and gives its users easy access to over 5,000 APIs as well as location information.

Pros:

1. One of its versions is available free of cost
2. Real-time fixes supported by the platform help to solve issues on the app itself
3. Includes easy drag-and-drop for several objects, which makes the app development task easy

Cons:

1. The open source version has limited utilities
2. Not a good option for development of complex mobile applications

Apache Cordova

Apache Cordova powers the build process for most of the free mobile app development tools. Adobe released Cordova as an open source project and so far it has received contributions from Google, IBM, BlackBerry, Microsoft and Intel. At first, Cordova bundles up CSS, JavaScript and HTML into a single client-side package and, subsequently, the software executes and renders the customised code within a native WebView. This is also known as the 'Hybrid application technique'. This approach supports 'Write-Once-Run-Anywhere' solutions. By wrapping a Web code into its native Cordova package, the platform can provide access to native APIs. By incorporating different community-built plugins, mobile applications can connect a number of those APIs just by using plain JavaScript.

Pros:

1. Developers working on Cordova have direct access to the latest updates from Apache's team
2. With each succeeding version, Cordova gains access to different critical OS enhancements
3. Cordova's performance improves with each of its subsequent enhancements and provides it access to new APIs

Cons:

1. UI and framework-agnostic
2. Dependent on developers for all design and architecture based issues

NativeScript

NativeScript is an open source mobile application development platform used by many of the industry veterans. It offers a similar software development experience as its backed-by-Facebook rival, React Native, but it resembles the Ionic framework development approach. NativeScript compiles the JavaScript mobile application to give a native mobile experience. It suggests Angular 2 as its preferred application development framework, but developers can

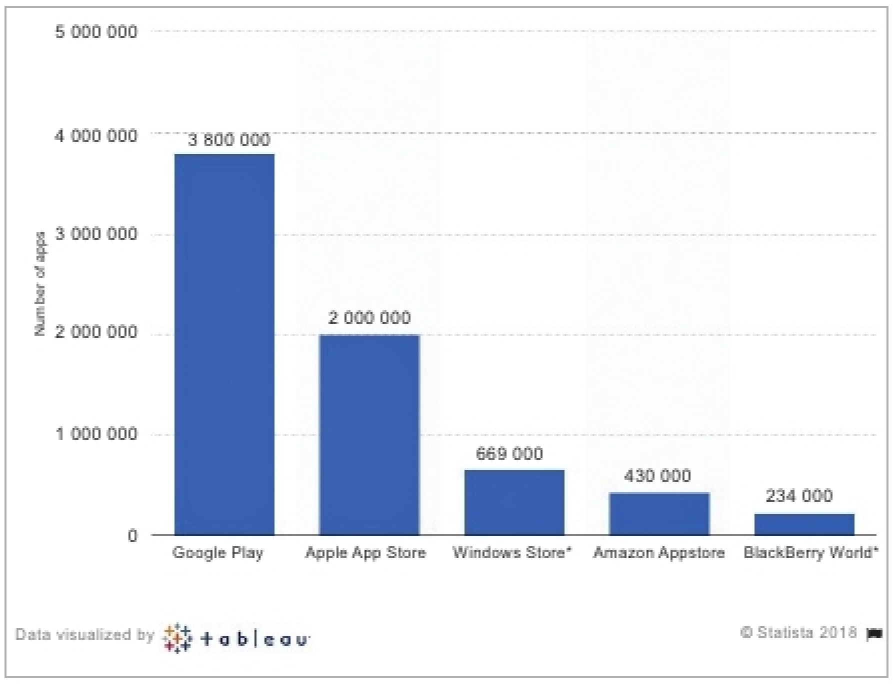

Figure 4: Number of mobile applications available on different mobile app stores (source: *Statista 2018*)

always opt out and choose the standard JavaScript with NativeScript APIs. It has Vue.JS integration and supports many of the available plugins for extended functionality.

In order to use NativeScript, knowledge of the command line is a must and we need to supply a text editor. NativeScript helps developers get a feel of the end product by providing an interface for demonstrating applications on iOS and Android. Both applications are generated from a single code base; hence NativeScript is a 'Write-Once-Run-Anywhere' solution capable of achieving very high performance on both platforms.

Pros:

1. Available completely free of cost as it is open source
2. Offers the same level of performance over hybrid solutions as well as when building native applications using JavaScript
3. Can be integrated with CocoaPods and Gradle directly, hence letting its developers incorporate Objective-C, native Swift and Java libraries into their NativeScript projects

Cons:

1. Low performance during high traffic
2. Uses a customised markup to design its interfaces without CSS3 or HTML5 END

References

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